

EduGenie Scalability and Future Plan

Current MVP Limitations

- No user authentication.
- No database storage.
- No saved learning history.
- No quiz score tracking.
- No PDF or DOCX upload.
- No voice assistant.
- No teacher dashboard.
- Gemini free-tier quotas may limit usage.

Short-Term Improvements

1. Database Integration

Add SQLite or PostgreSQL to store:

- Users
- Questions
- Quiz attempts
- Learning roadmaps
- Progress history

2. Authentication

Add:

- User registration
- Login
- Password protection
- User profiles

3. Quiz Scoring

Add:

- Answer selection
- Automatic scoring
- Score history
- Correct answer review

4. Document Upload

Support:

- PDF files
- Word documents

- Text files

The uploaded content can be extracted and summarized.

5. Learning Progress

Add:

- Completed topics
- Quiz scores
- Study streaks
- Progress charts
- Recommended next topics

Medium-Term Improvements

- Multilingual learning support
- Voice-based questions
- Teacher dashboard
- Course creation
- Collaborative study groups
- Personalized notifications
- Source citations
- Retrieval-augmented generation

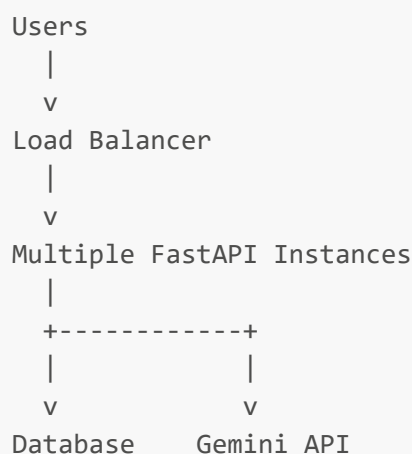
Deployment Plan

The application can be packaged using Docker.

Possible deployment targets:

- Cloud application hosting
- Virtual private server
- Container platform
- Educational institution server

Scalability Architecture



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Learning History and Progress